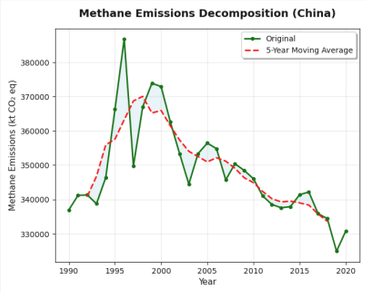
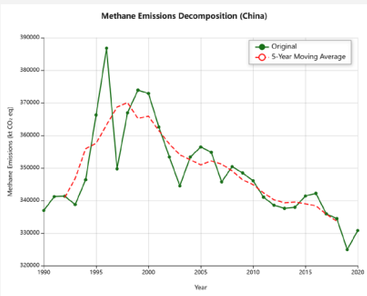
Original Figure

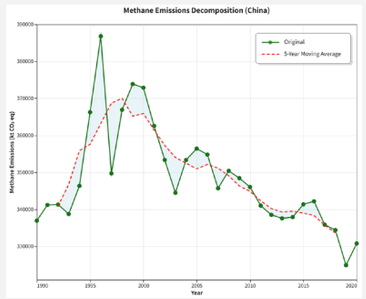
Python (Matplotlib/Seaborn)



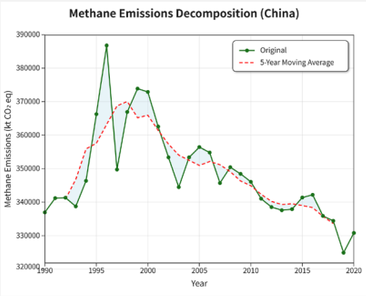
Apache ECharts



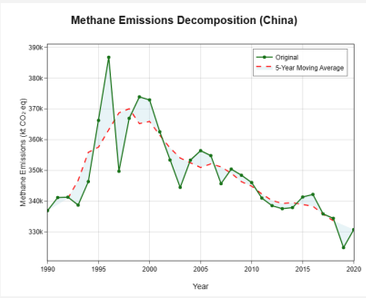
Vega-Lite



D3.js



Plotly.js



Key Code Changes

```
// gradient under Original
norm = Normalize(y.min(), y.max()); cmap = plt.cm.Greens
im = ax.imshow(np.linspace(ax.get_ylim()[0], y.max(), 256)[:, None],
           extent=[x.min(), x.max(), ax.get_ylim()[0], y.max()],
           origin='lower', aspect='auto', cmap=cmap, norm=norm, alpha=0.32,
zorder=0)
im.set_clip_path(Polygon(np.vstack([[x[0], ax.get_ylim()[0]], np.c_[x, y], [x[-1],
ax.get_ylim()[0]]]), closed=True, fc='none'))

// colorbar under legend (same cmap+norm)
legend = ax.legend(...)
bb =
legend.get_window_extent(fig.canvas.get_renderer()).transformed(ax.transAx
es.inverted())
fig.colorbar(ScalarMappable(norm=norm, cmap=cmap),
         cax=ax.inset_axes([bb.x0 + bb.width*0.39, bb.y0-0.34, bb.width*0.22,
0.32]),
         orientation='vertical')
```

```
// Original: vertical gradient area fill (bottom = lighter, top =
deeper)
areaStyle: { origin:'start', color:new
echarts.graphic.LinearGradient(0,1,0,0,[
  {offset:0,color:'rgba(0,100,0,0.08)'},{offset:1,color:'rgba(0,100
,0,0.45)'}
]) },


// Vertical colorbar under legend (min~max, same mapping:
low=light, high=deep)
visualMap: { type:'continuous', min:vMin, max:vMax,
orient:'vertical', right:75, top:140,
  inRange:{ color:['rgba(0,100,0,0.10)','rgba(0,100,0,0.55)'] }
}
```

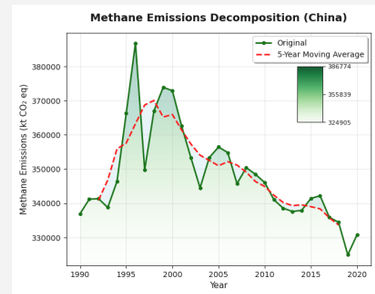
```
// Vertical gradient area (light bottom → dark top) filled to the x-axis
{ "mark":
{ "type":"area","opacity":0.22,"color":{ "gradient":"linear","x1":0,"x2":0,"
y1":1,"y2":0,
  "stops":[{"offset":0,"color":"#e5f5e0"},{"offset":1,"color":"#006d2c"}] } },
  "encoding":{ "x":{"field":"Year"},"y":{"field":"Original"},"y2":{"value":yDo
main[0]} } }


// Vertical colorbar as a single gradient rect
{ "mark":{ "type":"rect","color":{ "gradient":"linear","x1":0,"x2":0,"y1":1,"
y2":0,
  "stops":[{"offset":0,"color":"#e5f5e0"},{"offset":1,"color":"#006d2c"}] } },
  "encoding":{ "x":{"value":555},"x2":{"value":569},"y":{"value":100},"y2":{
"value":180} } }
```

```
// defs: vertical gradient
defs.append("linearGradient").attr("id","g").attr("x1","0%").attr(
"x2","0%").attr("y1","100%").attr("y2","0%")
  .selectAll("stop").data([{o:"0%",c:"#b8f5b8"},{o:"100%",c:"dark
green"}]).join("stop")
  .attr("offset",d=>d.o).attr("stop-color",d=>d.c);

// colorbar + labels
g.append("rect").attr("x",cbX).attr("y",cbY).attr("width",12).attr(
"height",85).attr("fill","url(#g)");
g.append("text").attr("x",cbX+18).attr("y",cbY+10).text(origMax);
g.append("text").attr("x",cbX+18).attr("y",cbY+85).text(origMin);
```

```
// Vertical gradient fill under Original
traceBoundary.fill = "tozeroy";
traceBoundary.fillgradient = { type:"vertical", colorscale:"Greens",
start:Math.min(...orig), stop:Math.max(...orig) };
traceBoundary.opacity = 0.35;

// Colorbar below legend (dummy invisible trace)
const traceColorbar = { x:[years[0]], y:[Math.min(...orig)], mode:"markers",
  marker:{ size:0, opacity:0, color:[Math.min(...orig),Math.max(...orig)],
   colorscale:"Greens", cmin:Math.min(...orig), cmax:Math.max(...orig),
showscale:true,
   colorbar:{ x:0.98, y:0.82, len:0.28, thickness:14,
tickvals:[Math.min(...orig),Math.max(...orig)] } },
  showlegend:false };
```

Updated Figure

Python (Matplotlib/Seaborn)



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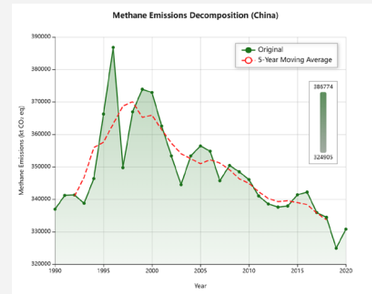

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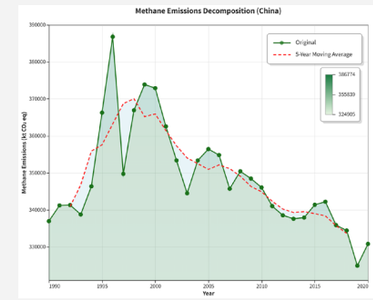
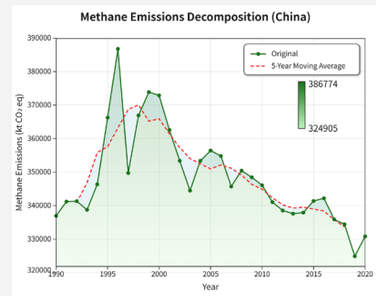
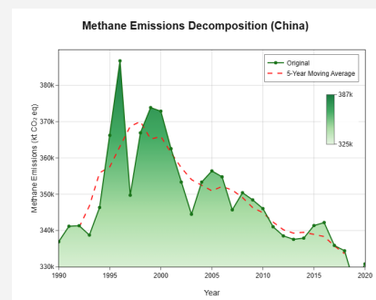


D3.js



Plotly.js